

Contract & Compliance Version-Intelligence Platform

Turning contract-review problems into
structured data-structure and algorithm
problems.

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CONTRACT

CLAUS

CLAUSE SEARCH

VERSI

VERSION DIFF

SIMIL

SIMILARITY

COMPL

COMPLIANCE

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Objectives

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From contracts
→ algorithms
→ outcomes



One platform. Five algorithmic jobs.

~~Large organisations manage thousands of contracts with repeated revisions. Manual compliance tracking is slow, error-prone, and difficult to audit.~~

Our platform applies classical data structures and algorithms to automate clause search, version diffing, similarity detection, obligation coverage, and reviewer assignment.

01

**CLAUSE
SEARCH**

Inverted index

02

**VERSION
DIFF**

Sequence alignment

03

**SIMILARITY
CLUSTER**

Graph methods

04

**OBLIGATION
COVERAGE**

Greedy Set Cover

05

**REVIEWER
ASSIGNMENT**

Bipartite matching

Why contract review becomes an algorithm problem⁰⁴

THE SCALE PROBLEM

- 1 **Thousands of contracts** Large corpus
- 2 **Repeated revisions** Many versions
- 3 **Clause-level changes** Hard to track
- 4 **Regulatory obligations** Coverage matters
- 5 **Limited reviewers** Workload matters

FIVE CORE ALGORITHMIC TASKS

- 1 **Clause indexing & search** Retrieve relevant clauses quickly
- 2 **Structured version diffing** Find added / removed / modified
- 3 **Similarity-based clustering** Discover related contracts
- 4 **Minimum Set Cover** Check obligation coverage
- 5 **Assignment algorithms** Allocate reviewers

Existing approaches solve pieces – our platform ~~connects them.~~

MANUAL REVIEW

Human side-by-side
checking

Slow and inconsistent at
scale

DOCUMENT COMPARISON

Raw document-level
diff

Not inherently clause-
aware

INFORMATION RETRIEVAL

Indexed text
search

Efficient retrieval at
scale

SEQUENCE ALIGNMENT

Structured version
comparison

Useful for revision
analysis

RESEARCH GAP

Individual techniques are commonly isolated. This project integrates indexing, alignment, graph/similarity, Set Cover and assignment into one auditable platform.

From manual audit burden to measurable algorithmic tasks

Manual contract auditing becomes inefficient when organisations have:

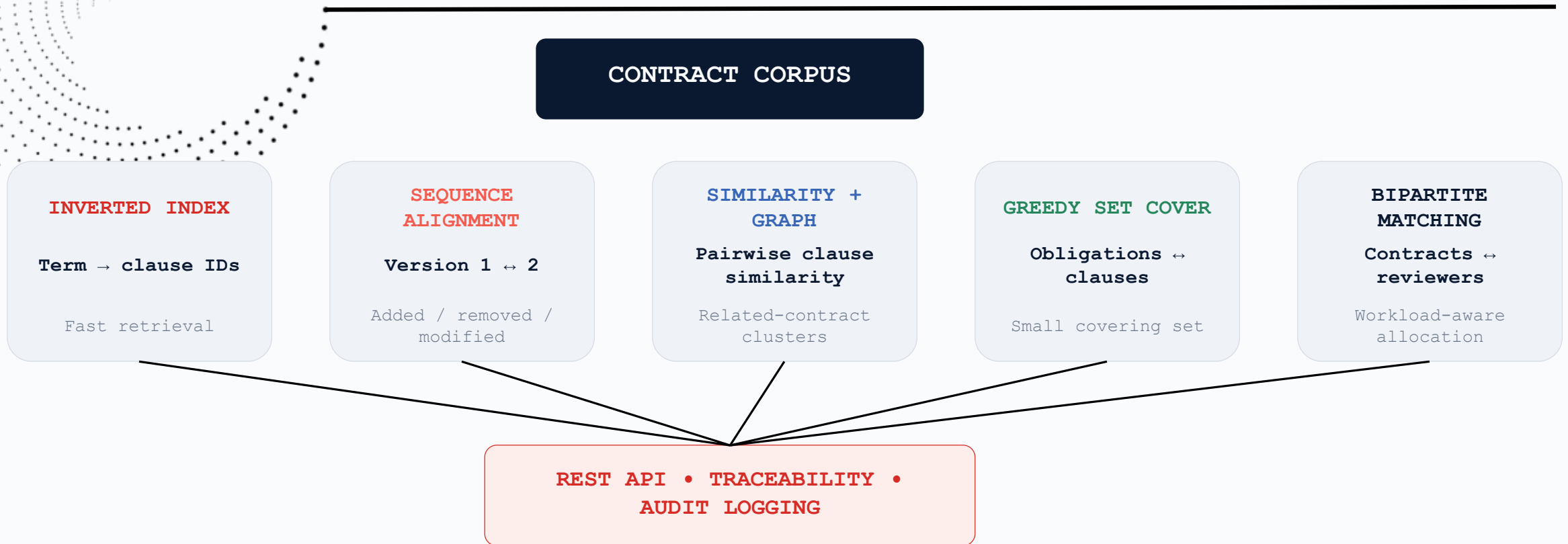
- large contract corpora
- repeated revisions
- regulatory obligations
- limited reviewer resources

SLOW • ERROR-PRONE • HARD TO AUDIT

OBJECTIVES

- ✓ Fast clause-level search across a corpus.
- ✓ Identify meaningful version changes at clause granularity.
- ✓ Discover relationships and clusters of similar contracts.
- ✓ Verify regulatory-obligation coverage systematically.
- ✓ Minimise clauses required to satisfy obligations (Set Cover).
- ✓ Assign reviewers considering workload and complexity.
- ✓ Deliver a unified, auditable API-based platform.

Algorithm workflow – the DSA core of the project



CORE FORMULATIONS Set Cover: minimize selected clauses subject to covering all required obligations • Matching: assign contracts to reviewers under workload/complexity constraints

Expected outcomes – what the platform produces

CLAUSE SEARCH

Fast retrieval

Relevant clauses
returned from the
indexed corpus

VERSION DIFF

Added • Removed • Modified

Structured comparison
between contract
versions

CONTRACT CLUSTERING

Related contracts grouped

Similarity relationships
represented as a graph

OBLIGATION COVERAGE

Small clause set

Coverage model
identifies clauses
satisfying obligations

REVIEWER ALLOCATION

Workload-balanced assignment

Matching model connects
contracts with reviewers

KEY TAKEAWAY

The result is not one algorithm – it is a coordinated pipeline where each DSA method solves a different contract-review bottleneck.

Technology stack & runtime

CORE RUNTIME

- Python 3.8+
- Google Colab

APP & ALGORITHMS

- FastAPI / Flask
- NetworkX
- difflib
- scikit-learn

DATA • TEST • VERSIONING

- SQLite3
- pytest
- Git / GitHub
- Windows 10/11 • Linux • macOS

Hardware: Google Colab cloud runtime; computational requirements depend on the selected runtime and corpus size.

A DSA solution to a real enterprise problem.

The platform integrates indexing, alignment, similarity analysis, Set Cover and assignment algorithms to automate core contract-review tasks.

VALUE

Faster review

More consistent
analysis

Auditable workflow

RETRIEVE

Inverted index

COMPARE

Sequence diff

RELATE

Similarity graph

COVER

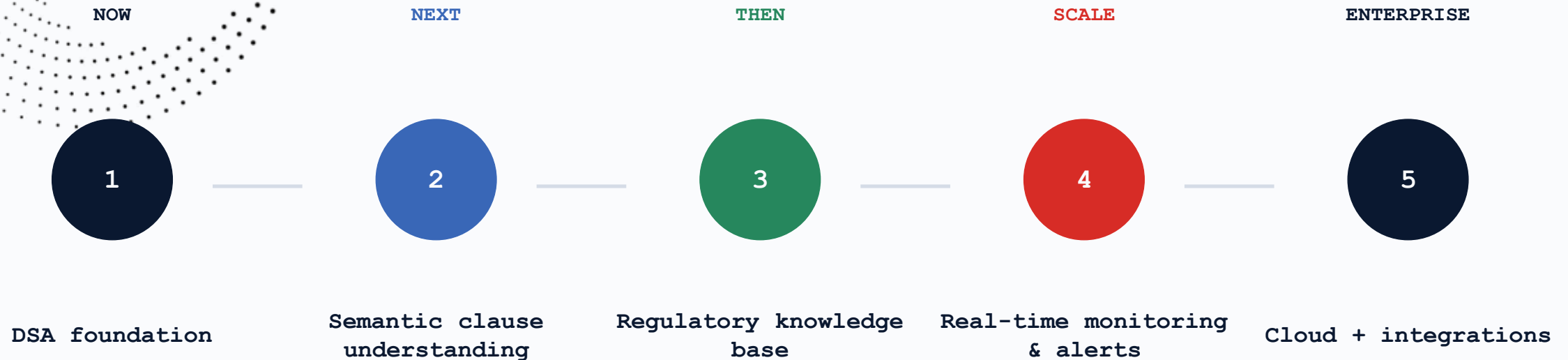
Set Cover

ASSIGN

Bipartite
matching

The architecture is modular, so future semantic/NLP capabilities can be added without replacing the DSA foundation.

Where the platform can go next



ADDITIONAL

Web reviewer dashboard • interactive diff & annotation • domain-adapted embeddings • automated periodic compliance verification & audit logs

Thank You

Questions?

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